

Printers, Then and Now

Inkjet printers penetrated the home segment only a few years back. Vendors such as HP and Canon developed inkjet-printing technology in the 1980s, but the prevailing prices were beyond the reach of most home users.

Working with these printers was also an expensive affair due to the cost of the ink used, the ink wasted during printing, and the overall expenditure involved in maintaining these printers. Dot matrix printers then were really slow and noisy and had many other disadvantages, but inkjet printers came into the scene and changed all that. With advancements in technology, these printers became increasingly efficient, and as sales volumes picked up, the prices too started steadily declining.

Today, most of these printers use either thermal or piezoelectric technology.

Thermal technology: This is the technology used in the Canon and Hewlett-Packard range of inkjet printers. Here, heat is used to fire ink onto the paper. The ink is actually squirted onto the paper, and this process is initiated by heating the ink to create a bubble until the pressure forces it to burst and hit the paper. The bubble then collapses as the ink cools, and the resulting vacuum draws ink from the reservoir to replace the ink that was ejected. Tiny heating elements are used to eject ink droplets from the print head's nozzles.

Piezoelectric technology: This is a proprietary Epson technology wherein a piezo crystal is placed at the back of the ink

reservoir. The piezo element oscillates when subjected to electrical voltage. To generate an ink drop, a current is applied to the piezo element. The element flexes and in doing so, it shoots a drop of ink out of the nozzle. The process helps to maintain control over the shape and size of the ink droplet. The tiny fluctuations in the crystal allow for smaller droplet sizes and hence higher nozzle density.

The speeds of printers using either of these two technologies have increased tremendously over the years. You can now choose from among various types of printers, and can also print in multiple modes (such as economical mode and best quality mode). So the next time you take an inkjet printout, think of the amazing technology that makes it possible.

sample letters, each having text combined with a graphic logo. The time taken to print out this sample file in black-and-white and gray scale modes was logged.

The image quality test consisted of two tests that were run using two separate sample prints. The first document was a combination of text, vector graphics and photos. The output of this document was evaluated to determine the printer's performance in terms of its ability to reproduce converging lines, concentric circles, high contrast text on a dark background, and the finer details in the original document.

To evaluate the performance of the printer in handling photographic prints, we used an actual A4-sized photograph that featured many variations in hue and brightness, and also had many details and

fine nuances.

We also considered features such as build quality, thoroughness of the documentation, bundled software, etc, with points being awarded for each feature.

Based on the price, two categories emerged: budget printers priced below Rs 15,000, intended for home use, and mid-range printers costing more than Rs 15,000, used more in offices.

Speed

For comparing the speed of the printers, the text document was printed in normal draft mode (or Lexmark Z52: Sturdy and fast black-and-white mode) on plain paper, and then again using gray scale mode. We then printed the image document in both gray scale and normal colour modes on plain paper. We logged the time taken for each of these printouts to determine the printer's performance in each of the two modes.

The **Lexmark Z52** clocked the fastest time of 36 seconds for printing the five-page text document in normal mode, followed by Hewlett-Packard's DJ 990Cxi (42 seconds) and the Epson Stylus Color 880 (60 seconds). So if the bulk of your printing applications at home or office involves text documents, the Lexmark Z52, with its combination of speed and excellent print quality, is the best choice.

It printed the text document in gray scale mode in 55 seconds, which was the fastest time logged in this mode. The runners-up (in gray scale mode) were the Lexmark Z42 (61 seconds) and the Canon BJC 2100SP (88 seconds).

You may well wonder why you would ever need to print a document in gray scale mode. Well, if you have used colours in a document and just need a printout to proof it, you could take the printout in gray scale mode and save your expensive colour cartridge for the final printouts.

In the image document test in normal mode, the winner was the **Epson Stylus Color 880** with a print speed of 31 seconds, followed by the Canon BJC 5100 (43 seconds) and the Epson Stylus Color 680 (62 seconds). If you need a printer to execute volume-printing tasks in colour, go for the Epson Stylus Color 880.

In the final speed test, wherein the image document was printed in gray scale mode, the **Epson Stylus Color 680** emerged as the winner with a timing of 44 seconds. Three printers came in a close second—the Lexmark Z52, the Epson Stylus Color 880, and the Epson Stylus Photo 870, with all three clocking 63 seconds. So if you regularly proof images and want to conserve your colour cartridge, you have a choice of four printers.

